

# Chancharik Mitra

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## EDUCATION

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### Carnegie Mellon University

*M.S. in Machine Learning GPA: 3.95/4.00*

Pittsburgh, PA, USA

*Aug. 2024 – Present*

### University of California, Berkeley

*B.S. in EECS - Research Honors; GPA: 3.87/4.00*

Berkeley, CA, USA

*Aug. 2021 – May 2024*

### Moon Area High School

*High School Diploma; GPA weighted: 4.70/4.00 GPA unweighted: 4.00/4.00*

Pittsburgh, PA, USA

*Aug. 2017 – Jun. 2021*

*College Courses during High School: Harvard University, University of Pittsburgh, and Robert Morris University - all 4.00 GPA*

## RESEARCH INTERESTS

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My research aims to create AI systems that leverage pretrained knowledge to efficiently adapt to new tasks and environments. I pursue two core directions: (1) few-shot adaptation via mechanistic interpretability, targeting task-relevant representations without expensive fine-tuning, and (2) systematic generalization through cognitive and linguistic frameworks such as compositionality and pragmatics. I apply these methods to vision-language-action applications such as video reasoning, robotics, and education.

**See research overview figure:** <https://chancharikmitra.github.io/media/research-overview.png>

## RESEARCH EXPERIENCE

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### Carnegie Mellon University - Robotics Institute

Pittsburgh, PA

*Research Assistant - Advised by Dr. Deva Ramanan*

*May 2024 – Present*

- Conducting research on *few-shot mechanistic interpretability for model adaptation*, leveraging activation steering and feature extraction to enable **efficient adaptation of VLMs and VLAs** from limited examples across vision-language **classification**, multimodal **reward modeling**, **object detection**, and **robotic applications**
- Developing evaluation methods for spatiotemporal compositionality in video-language models, examining event properties such as object-attribute-relationship state transitions, event order, and speed
- Advancing compositional **video understanding and generation** through camera motion research: developed CameraBench and finetuned SOTA LMM for camera understanding; currently working on compositional cinematic video captioning and generation with fine-grained camera control

### Berkeley AI Research (BAIR) Lab

Berkeley, CA

*Researcher - Advised by Dr. Trevor Darrell and Dr. Dan Klein*

*Apr. 2022 – August 2025*

- Advanced *few-shot mechanistic interpretability for model adaptation* through task vectors, activation steering, and feature extraction methods for Large Multimodal Models (LMMs), demonstrating superior performance over baselines including LoRA
- Developed methods for compositional vision-language understanding, leveraging scene graphs and chain-of-thought prompting to enhance multimodal reasoning capabilities
- Applied pragmatic reasoning principles to improve contextual language grounding in embodied settings for 3D object disambiguation
- Contributions **featured in IBM's Granite Vision models and AllenAI's MOLMO**

### University of California, Berkeley – EECS Department

Berkeley, CA

*Research Assistant - Advised by Dr. Gireeja Ranade and Dr. Narges Norouzi*

*May 2023 – August 2025*

- Conducting research on multimodal question-answering and retrieval-augmented generation for educational AI applications
- *Edison*: Deployed LLM-based QA system in production for UC Berkeley's DATA 100 course (~1000 students), with document retrieval pipeline and pedagogically-informed prompting strategies
- *Askademia*: Developed multi-institutional educational QA platform with advanced retrieval methods and evaluation frameworks for assessing pedagogical value

## National Center for Supercomputing Applications (NCSA @ UIUC)

Champaign, IL

*Undergraduate Researcher - Advised by Dr. Aiman Soliman and Dr. Zeynep Madak-Erdogan Apr. 2022 – Aug. 2024*

- Conducted research as an REU fellow and undergraduate researcher (post-program) on leveraging machine learning techniques to segment spatial transcriptomics data of ER+ breast cancer tissue samples
- First-authored paper published in the *IEEE MASS '22* conference and gave an oral presentation of the research at REUNS workshop (won **Best Paper** award) as well as at the Illinois Summer Research Symposium (ISRS)
- Developed an explainable, model-agnostic biomarker mining pipeline for spatial transcriptomics data

## WORK EXPERIENCE

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### Carnegie Mellon University – Robotics Institute

Pittsburgh, PA

*Research Assistant - Advised by Dr. Deva Ramanan*

*May 2025 – August 2025*

- Conducting research on generalizable vision-language-action reasoning via few-shot mechanistic interpretability for model adaptation and compositional video understanding and generation
- For detailed research directions and related publications, please refer to the corresponding research section entry

### Carnegie Mellon University – Machine Learning Department

Pittsburgh, PA

*10-718 Machine Learning in Practice: Teaching Assistant*

*Aug. 2025 – Present*

- Teaching assistant for graduate course covering ML specification, evaluation, robustness, interpretability, and other concepts essential for successful research and industry ML projects
- Responsibilities include hosting office hours, grading assignments, lecture support, and exam question development

### PromptAI

Berkeley, CA

*Research Intern*

*Jun. 2024 – Sept. 2024*

- Developed an open-source video LMM model using InternVideo2 video encoder and LLaMA3 language model
- Created video-image in-context learning dataset using Panda-70M and OBELICS dataset to enable video in-context learning (ICL) capabilities
- Investigated the applications of video ICL and multimodal task vectors for open-source models, demonstrating that VILA-1.5 and LLaVA-OneVision can surpass GPT-4o on visual tasks using these methods

### University of California, Berkeley – EECS Department

Berkeley, CA

*EECS 16B Designing Information Devices and Systems II: Lead Discussion and Content TA Jun. 2022 – May 2024*

- Taught weekly discussion sections of over 50 students
- Developed discussion, midterm, and final exam problems for topics ranging from DC/AC circuits, differential equations, linear algebra, control, SVD/PCA, and classification
- For samples of my content creation, please refer to any discussion or exam on the [EECS 16B course page](#)
- Restructured course to emphasize circuit, Bode Plot, and SVD/PCA concepts

### National Center for Supercomputing Applications (NCSA @ UIUC)

Champaign, IL

*REU FoDOMMaT Undergraduate Research Fellow*

*Jun. 2022 – Aug. 2023*

- Accepted to the selective REU-FoDOMMaT summer research program held at NCSA on-campus at the University of Illinois, Urbana-Champaign (UIUC). Invited to continue research post-program and to return as a Research Fellow the following summer.
- For more details, please refer to the corresponding research section entry.

## PUBLICATIONS

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*More information (links, project pages, etc.) available on my website: <https://chancharikmitra.github.io/>*

### Submitted (In Review):

- [1] [Activation Reward Models for Few-Shot Model Alignment](#). Tianning Chai\*, Chancharik Mitra\*, Brandon Huang, Gautam Rajendrakumar Gare, Zhiqiu Lin, Assaf Arbelle, Leonid Karlinsky, Rogerio Feris, Trevor Darrell, Deva Ramanan, and Roei Herzig. In: *Annual Meeting of the Association for Computational Linguistics (ACL)*. 2026. Submitted.
- [2] [Building a Precise Video Language with Human-AI Oversight](#). Chancharik Mitra\*, Siyuan Cen\*, Zhiqiu Lin\*, Isaac Li, Yuhan Huang, Yu Tong Tiffany Ling, Hewei Wang, Irene Pi, Shihang Zhu, Yili Han, Yilun Du, and Deva Ramanan. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2026. Submitted.

- [3] [Mechanistic Finetuning of Vision-Language-Action Models via Few-Shot Demonstrations](#). Chancharik Mitra\*, Yusen Luo\*, Raj Saravanan\*, Dantong Niu, Anirudh Pai, Jesse Thomason, Trevor Darrell, Abrar Anwar, Deva Ramanan, and Roei Herzig. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2026. Submitted.

**Published:**

- [4] [CameraBench: Towards Understanding Camera Motions in Any Video](#). Zhiqiu Lin\*, Siyuan Cen\*, Daniel Jiang, Jay Karhade, Hewei Wang, Chancharik Mitra, Tiffany Ling, Yuhao Huang, Sifan Liu, Mingyu Chen, et al. In: *Conference on Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*. 2025. **Spotlight**.
- [5] [Enhancing Few-Shot Vision-Language Classification with Sparse Attention Vectors](#). Chancharik Mitra, Brandon Huang, Tianning Chai, Zhiqiu Lin, Assaf Arbelle, Rogerio Feris, Leonid Karlinsky, Trevor Darrell, Deva Ramanan, and Roei Herzig. In: *International Conference on Computer Vision (ICCV)*. 2025.
- [6] [Analyzing Pedagogical Quality and Efficiency of LLM Responses with TA Feedback to Live Student Questions](#). Chancharik Mitra\*, Mihran Miroyan\*, Rishi Jain\*, Gireeja Ranade, and Narges Norouzi. In: *ACM Technical Symposium on Computer Science Education (SIGCSE)*. 2025.
- [7] [EduMod-LLM: A Modular Approach for Designing Flexible and Transparent Educational Assistants](#). Meenakshi Mittal, Rishi Khare, Mihran Miroyan, Chancharik Mitra, and Narges Norouzi. In: *AAAI Conference on Artificial Intelligence*. 2025.
- [8] [Raising the Bar: Automating Consistent and Equitable Student Support with LLMs](#). Meenakshi Mittal\*, Azalea Bailey\*, Victoria Phelps\*, Rose Niousha, Mihran Miroyan, Chancharik Mitra, Rishi Jain, Gireeja Ranade, and Narges Norouzi. In: *ACM Technical Symposium on Computer Science Education (SIGCSE) Poster*. 2025.
- [9] [Less is More Tokens: Efficient Math Reasoning via Difficulty-Aware Chain-of-Thought Distillation](#). Abdul Waheed\*, Chancharik Mitra\*, Laurie Z. Wang\*, Deva Ramanan, and Bhiksha Raj. In: *NeurIPS Workshop on Multimodal Algorithmic Reasoning*. 2025.
- [10] [Multimodal Task Vectors Enable Many-Shot Multimodal In-Context Learning](#). Brandon Huang\*, Chancharik Mitra\*, Assaf Arbelle, Leonid Karlinsky, Trevor Darrell, and Roei Herzig. In: *Conference on Neural Information Processing Systems (NeurIPS)*. 2024.
- [11] [Compositional Chain-of-Thought Prompting for Large Multimodal Models](#). Chancharik Mitra, Brandon Huang, Trevor Darrell, and Roei Herzig. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2024.
- [12] [Which One? Leveraging Context Between Objects and Multiple Views for Language Grounding](#). Chancharik Mitra\*, Abrar Anwar\*, Rodolfo Corona, Dan Klein, Trevor Darrell, and Jesse Thomason. In: *North American Chapter of the Association for Computational Linguistics (NAACL)*. 2024.
- [13] [Elevating Learning Experiences: Leveraging Large Language Models as Student-Facing Assistants in Discussion Forums](#). Chancharik Mitra\*, Mihran Miroyan\*, Rishi Jain\*, Vedant Kumud, Gireeja Ranade, and Narges Norouzi. In: *ACM Technical Symposium on Computer Science Education (SIGCSE) Poster*. 2024.
- [14] [RetLLM-E: Retrieval-Prompt Strategy for Question-Answering on Student Discussion Forums](#). Chancharik Mitra\*, Mihran Miroyan\*, Rishi Jain\*, Vedant Kumud, Gireeja Ranade, and Narges Norouzi. In: *AAAI Conference on Artificial Intelligence*. 2024.
- [15] [Geospatial Analysis To Quantify Spatial Heterogeneity Of Tumor Microenvironment](#). Jin Young Yoo, Chancharik Mitra, Yang Yue, Aiman Soliman, and Zeynep Madak-Erdogan. *Journal of the Endocrine Society*. 2023.
- [16] [Spatial Analysis of Tumor Heterogeneity Using Machine Learning Techniques](#). Chancharik Mitra, Jin Young Yoo, Zeynep Madak-Erdogan, and Aiman Soliman. In: *IEEE International Conference on Mobile Ad Hoc and Smart Systems (MASS)*. 2022. **Best Paper Award** at REUNS Workshop.

## AWARDS & ACHIEVEMENTS

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**NeurIPS Datasets and Benchmarks Spotlight for *CameraBench*** | 2025

**National Science Foundation Graduate Research Fellowship** | 2025

**UC Berkeley “Outstanding Graduate Student Instructor Award”** | 2024

**Berkeley EECS Research Honors Program Recipient** | 2023

**Google CS Research Mentorship Program Recipient** | 2023

**2x UC Berkeley College of Engineering Dean’s List Award** | top 10% semester GPA - 2022, 2023

**Best Paper Award** | Awarded “Best Paper” at the *8th National Workshop for REU Research in Networking and Systems* colocated with the *19th IEEE International Conference on Mobile Ad-Hoc and Smart Systems* - 2022

**USACO Gold** | Achieved a perfect score on USA Computing Olympiad Bronze and Silver division competitions, earning Gold-level USACO standing - 2020/2021

**2x American Invitational Mathematics Exam (AIME) Qualifier** | Earned a top 5% score nationally on the American Mathematics Competition 12 (AMC 12) exam to qualify for the AIME exam, going on to score in the top 20% of AIME qualifiers - 2020/2021

## PROJECTS

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**All About Autoencoders** | [GitHub](#)

- Comprehensive theoretical and practical survey of autoencoder architectures for Berkeley’s CS 182 Deep Learning
- Covers vanilla, variational, and vector-quantized VAE implementations, theoretical exercises, and illustrative visualizations including latent space exploration, denoising, and ablation studies
- Received full marks plus extra credit for exceeding project requirements
- Resource incorporated into a [comprehensive machine learning course](#) at UIUC

## VOLUNTEERING & STUDENT ORGANIZATIONS

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### Computer Science Mentors

Berkeley, CA

*Senior Teaching Mentor*

*Jun. 2022 – May 2024*

- Volunteer Senior Mentor for CS 70 (Discrete Mathematics & Probability) – STEM Excellence through Equity and Diversity (SEED) Scholars section
- Responsibilities include teaching a section of students, leading a group of junior mentors, and contributing to CS 70’s content task force: recording video walkthroughs of problems as well as creating exam review material

### IEEE – Berkeley Student Branch

Berkeley, CA

*Technical Operations Officer*

*Jun. 2022 – August 2023*

- Responsible for maintaining IEEE’s server which houses the organization’s website and technical project resources

### STEM Coding Lab

Pittsburgh, PA

*Volunteer Coding Instructor and Curriculum Developer*

*Aug. 2018 – June 2022*

- STEM Coding Lab is a nonprofit organization with the mission of teaching children from underserved communities in Pittsburgh how to code
- Teach programming classes (e.g. Python, Java, HTML/CSS/Javascript) of 20+ students (grades K-12) multiple times a week
- Developed an 8-week Python course that was used to teach students at the Pittsburgh Public Schools during the COVID-19 pandemic

## SKILLS

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**Programming:** Python, Pytorch, NumPy, Scikit Learn, Java, C, Assembly, Tensorflow, SQL

**Technologies and Software:** Git, Linux, Windows, MacOS, Arduino, Tableau, Android Studio

**Languages:** English (Native), Bengali (Native), Hindi (Advanced), German (Intermediate)

## RELEVANT COURSEWORK

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**CMU:** Computer Vision (16-720) - Grade A+, Advanced Natural Language Processing (11-711) - Grade A+, Machine Learning in Practice - Grade A+

**Berkeley:** CS 288 Graduate-level Natural Language Processing - Grade A, PLANTBI 200B Graduate-level Genomics and Computational Biology - Grade A, CS 182 Deep Learning (completed graduate course requirements for extra credit) - Grade A, CS 188 Artificial Intelligence - Grade A+, CS 189 Machine Learning - Grade A, MCELLBI 132 Biology of Human Cancer - Grade A, EECS 16B Designing Information Devices and Systems I (Linear Algebra, Differential Equations, and Circuits) - Grade A+